Northeastern University
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Prof: Andrei Guschin
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Assignment - MRD Report

List of all students who actively worked on this MRD Report

Aditya Girish Anand
Matthieu Delatour
Dorian Chirackal
Karishma Bahl
Sangeetha Phaneesh

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Vision

For patients and families who struggle to manage medical records scattered across different hospitals, clinics, and doctors, MediVault is a unified healthcare app that brings all records together in one secure place, fully owned and controlled by the patient. Unlike hospital-specific portals or paper-based files, our product includes an AI assistant that listens during doctor visits, summarizes prescriptions, and keeps a lifelong consultation history, giving both patients and doctors quick access to complete medical information for faster and more efficient care. We make sure that our emphasis on HIPAA-compliant security and patient autonomy is never compromised by making the user the sole owner and curator of their health history. We distinguish ourselves from competitors like MyChart, FollowMyHealth, and traditional EHR portals because of our distinctive emphasis on unification, control, and the AI-driven consolidation of new data. Our approach aims to fill the gap left by those who need to maintain a single, comprehensive, and portable health record but lack the tools to bypass complex, proprietary systems.

Motivation

1. Customer Segments

i. Segment A – Young Adults Managing Complex Mental-Health Care

Description

Adults 18-25 years old diagnosed with anxiety or depression, actively seeing two or more mental health providers per year (e.g., therapist, psychiatrist, and primary-care physician). They typically manage at least one prescription (e.g., SSRI, anxiolytic) and face frequent adjustments or documentation gaps between providers.

Pain Points

- Repeating history and medication list at every appointment.
- Difficulty reconciling changes across different EHR systems.
- Lack of visibility into therapy progress or prescription history.

Adoption Drivers

- High smartphone ownership (≈ 96% among adults aged 18-25).[i]
- Post-pandemic normalization of digital mental health tools (BetterHelp, Talkspace).
- High prevalence of mental illness among adults aged 18-25 (≈31% any mental illness, 2023[ii].
- Significant overlap of multiple provider visits (~40% of treated adults [iii]).

User Persona

Jordan Rivera

Age: 24

Occupation: Graduate Student in Law

Location: Boston, MA

Education: JD Candidate at Northeastern University

Family: Lives with partner; sees therapist and psychiatrist

Behaviors & Tools:
- Tracks therapy progress across notes, phone apps, and clinic portals
- Uses university counseling systems that do not integrate with outside providers
- Sometimes forgetting past medications due to scattered data

Pain Points:
- Frustrated retelling mental-health history every time a new provider is involved
- Lost notes or inaccessible records during provider transitions
- Feels emotionally drained repeating sensitive information

Trust & Security Concerns:
- Deep concern about confidentiality of therapy notes
- Willing to trust only verified or university-linked systems
- Needs ability to delete and control access personally

Key Opportunity for MediVault:
MediVault can gain trust through:
- End-to-end encryption and granular permissions (user controls who sees what).
- Provider-verified integrations (e.g., university or clinic portals).
- Easy record export/import for changing therapists or psychiatrists.

Representative Quote:
"My story is personal. I'll use tech if I can decide who sees what. Not the other way around."

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ii. Segment B - Parents of Infants (0-2 Years)

<u>Description</u>
Parents aged 22-40 managing pediatric vaccination schedules, growth charts, and health records for infants and toddlers under age 2.

<u>Pain Points</u>
- Paper-based vaccination cards are easily lost (common issue noted by AAP [iv]).
- Lack of unified digital access to pediatric visit summaries.
- Coordination issues when changing pediatricians or day-care providers.

<u>Adoption Drivers</u>
- Smartphone usage among parents under 40 exceeds 80% [v].
- Strong digital parenting culture (apps like BabyCenter, What to Expect).
- Clear utility need - CDC recommends 10+ pediatric visits before age two [vi].
- ~3.6 million annual U.S. births [vii] ensure a constantly renewing user base.

<u>User Persona</u>
Melissa Chang
Age: 33
Occupation: Product Manager
Location: Somerville, MA

Family: Married, mother of a 2-year-old and expecting second child
Education: MBA in Business Analytics

Behaviors & Tools:
- Tracks vaccinations in paper folders and digital photos
- Uses messaging apps to share records with spouse and daycare
- Relies on pediatric clinic portal but finds it incomplete

Pain Points:
- Lost or misplaced vaccine card delayed daycare registration
- Missed vaccination reminder caused schedule confusion
- Duplicate information across paper, email, and hospital portals
- Frustration when specialists request documents she cannot find

Trust & Security Concerns:
- Strong need for parental control and consent before data sharing
- Accepts digital systems only if encrypted and verified by medical institutions
- Wants transparency about data access logs

Key Opportunity for MediVault:
MediVault can offer peace of mind through:
- Digital vaccination passport linked to pediatric EMRs.
- Two-factor authentication and visible access history.
- Parental sharing controls for daycare and healthcare providers.

Representative Quote:

"If I can control who sees my child's records and it's verified by my pediatrician — I'd switch in a heartbeat."

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iii. Segment C - Seniors with Polypharmacy & Caregiver Support

Description
Adults 65+ taking five or more prescriptions (polypharmacy threshold) and relying on at least one informal caregiver for appointment coordination.

Pain Points
- Medication duplication or omission due to fragmented records.
- Caregivers lack easy, HIPAA-compliant access to patient histories.
- Difficulty keeping multiple specialists aligned (cardiology, endocrinology, etc.).

Adoption Drivers
- 61% smartphone ownership among seniors [viii].
- 38% of adults aged 65+ take five or more prescriptions [ix].
- 53 million unpaid caregivers in the U.S. [x].
- 58 million adults aged 65+ in 2024 [xi].
- Regulatory support: Medicare interoperability mandates (2023+) [xii].

User Persona

Eleanor Harris

Age: 71
Occupation: Retired Librarian
Location: Brookline, MA
Family: Widowed; daughter assists remotely with appointments
Education: MA in History

Behaviors & Tools:

- Keeps prescriptions written in a small notebook and folders for each specialist visit
- Often relies on handwritten medication lists and paper discharge notes
- Uses her daughter's help to coordinate prescriptions and appointments

Pain Points:

- Nearly received conflicting medication due to missing list
- Frustrated when hospitals do not share results, forcing repeated tests
- Difficulty finding latest reports during emergencies
- Stressful medication adjustments when switching doctors

Trust & Security Concerns:

1. Concerned about hackers and ads in health apps
2. Wants to control which doctors have access
3. Prefers offline or hospital-approved apps

Key Opportunity for MediVault:

MediVault can help Eleanor by offering:

- A secure, offline-accessible record system for medications and reports.
- Caregiver access management, allowing her daughter to support coordination.
- Hospital-linked encryption to increase trust.

Representative Quote:

"I don't want another password nightmare, just a safe place where my daughter and doctor can both see what I take."

2. Unmet Needs

We believe that each of our target segments has unique unmet needs. Thus, we pinpointed the problems faced by our primary 3 segments and mentioned them below per demographic.

i. Young and Middle-aged Adults Managing Complex Mental-Health Care

- Continuity of care across multiple providers (therapist, psychiatrist, primary care) without having to retell their complete mental health history at every appointment.
- Granular privacy controls allow them to decide exactly what information to share with which providers, ensuring confidentiality of sensitive therapy notes.
- Seamless integration between university counseling systems and outside mental health providers to prevent information loss during provider transitions.
- Digital record accessibility that reduces the emotional exhaustion of repeatedly sharing personal trauma and medication history.

ii. Parents of Infants (0-2 Years)

- Secure digital vaccination records that eliminate the risk of lost paper cards and prevent daycare enrollment delays or rejection.
- Unified record system that consolidates vaccination schedules, growth charts, and pediatric visit notes in one accessible place instead of scattered across paper, emails, and hospital portals.
- Automated reminders for upcoming vaccinations and pediatric appointments to prevent missed schedules and confusion.
- Easy sharing controls allow parents to provide daycare, schools, and specialists with verified medical records without running home for physical documents.

iii. Seniors with Polypharmacy & Caregiver Support

- Centralized medication management that prevents dangerous drug interactions when visiting multiple specialists who do not share prescription information
- HIPAA-compliant caregiver access that allows family members to coordinate appointments and manage prescriptions without complex technical barriers.
- Offline-accessible records with large-font interfaces that accommodate limited technical literacy and ensure medication lists are available during emergencies.
- Simplified authentication systems that do not create "password nightmares" while maintaining hospital-grade security and encryption.
- Automatic record synchronization across specialists to eliminate repeated tests and ensure all doctors have complete patient histories.
- Research Design
  We conducted 20 in-depth customer interviews (7 mental health students, 6 parents of infants, 7 seniors/caregivers) and 10 provider/insurer stakeholder interviews.
  Additionally, we surveyed 20 potential users with structured questions about record-keeping methods (Q1), specific problem incidents (Q2), and trust factors (Q3).

The consistency of pain points across interviews (83-100% reporting similar issues within each segment) provides strong validation of these unmet needs.

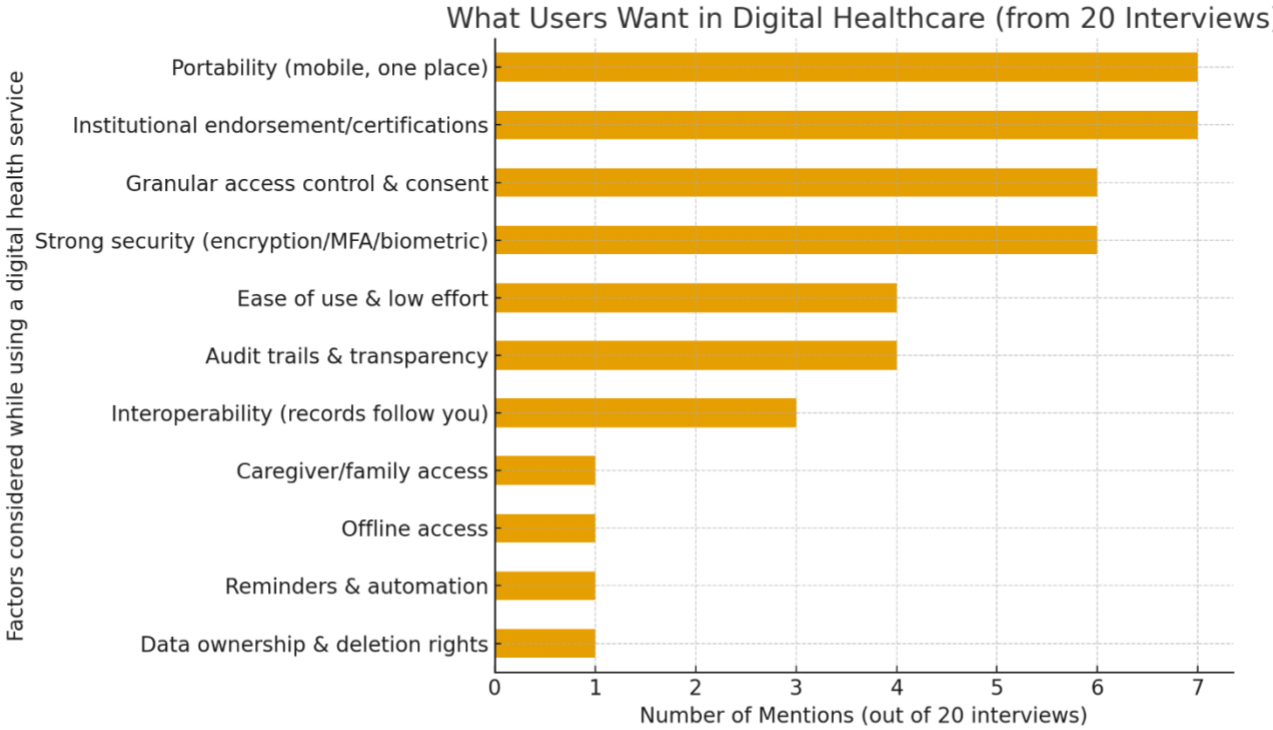

Figure 1 - What users want in a Digital Healthcare solution? (based on users interviews feedback (see Appendix)

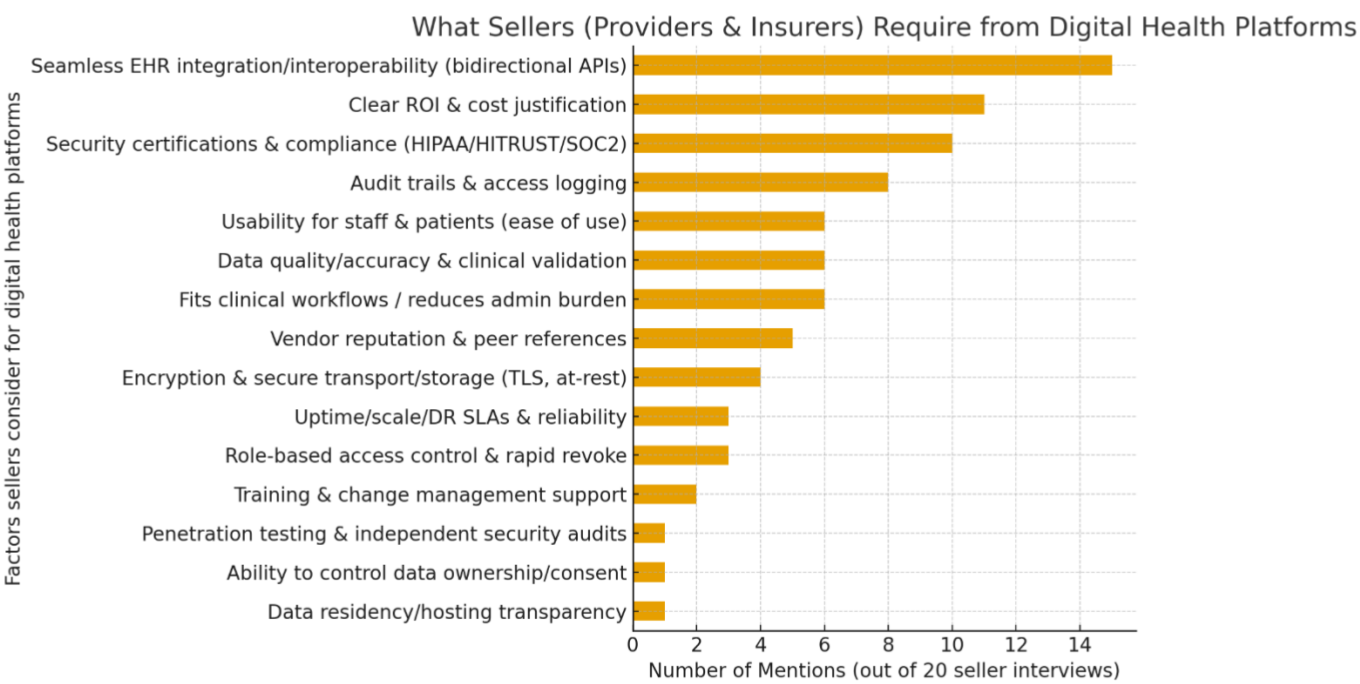

Figure 2 - What sellers want from a Digital Healthcare solution? (based on users interviews feedback (see Appendix)

3. Existing Solutions

The healthcare technology ecosystem features a wide range of digital platforms designed to enhance patient engagement, data accessibility, and clinical documentation. However, despite the abundance of tools, true interoperability and comprehensive patient record consolidation remain limited.
Current solutions can be divided into two main categories:

1. Patient-Facing Portals and Health Apps, such as MyChart*,* FollowMyHealth*,* Apple Health*,* eClinicalWorks Patient Portal*,* and OneRecord, which aim to give patients visibility into their medical data, test results, and appointment history. While these solutions have improved access to information, they often operate within closed or semi-closed networks, restricting the patient's ability to view all records across multiple providers or systems.
2. Provider-Facing AI Documentation Tools, such as *Nuance DAX* and *Sunoh.ai*, which focus on automating note-taking and clinical documentation. These solutions enhance efficiency for healthcare providers but do not address the patient's challenge of having fragmented health records spread across different institutions.

Collectively, these existing platforms highlight a critical gap: a unified, patient-centric solution that aggregates data seamlessly from multiple providers and systems, offering patients a comprehensive, portable health record under their control.

(1) MyChart is a widely used patient portal developed by Epic Systems. It allows patients to view test results, schedule appointments, message doctors, and review their medical history for hospitals and providers that use Epic's EHR. However, it is limited to providers within the Epic ecosystem, so if a patient sees providers on other systems, MyChart may not include all records.

(2) FollowMyHealth is a third-party patient portal offered by Allscripts that is used by a variety of healthcare organizations. It offers functions such as viewing test results and messaging providers. Yet, its limitation is that each provider must be part of the FollowMyHealth network for integration and full data sharing.

(3) Apple Health (Health Records feature): This is a consumer-facing app by Apple which consolidates health data (medications, labs, immunizations) from participating devices, apps and hospitals. It allows patients to connect their iPhone to participating health systems to pull in some records. However, its scope is limited and depends on each hospital or clinic providing API connections and enabling the service.

(4) Nuance DAX (Dragon Ambient Experience): Developed by Nuance Communications, this is an AI-powered ambient scribe tool that listens during patient-physician conversations and automates clinical documentation for the provider side. It is not a patient-facing record consolidation platform, but a provider's workflow tool.

(5) eClinicalWorks Patient Portal offered by eClinicalWorks is a portal that supports independent practices and provides access to medical summaries, secure messaging, and basic record-viewing. It does not unify records from disparate systems outside that network, so fragmentation remains a limitation.

(6) OneRecord is a patient-facing app that lets users access, aggregate, and share their full medical history across multiple providers and EHR vendors. It emphasizes ownership by patients and supports integration across systems (Epic, Cerner, Allscripts, etc.). Its limitation: the user typically must manually connect each provider account, and full data completeness depends on provider participation in the network.

(7) Sunoh.ai is an AI medical-scribe solution that supports providers by turning patient-provider conversations into clinical notes and integrates with EHRs. It is not designed primarily as a patient record consolidation platform; its focus is on documenting visits rather than giving full patient-controlled aggregated records.

Feature/ Product	MyChart	FollowMy Health	Apple Health	Nuance DAX	eClinical WorkPortal	Sunoh.ai	One Record
Price	Free	Free	Free	Enterprise license	Free	$149/mo (provider)	Free
Target Users	Epic patients	Network patients	General Consumers	Clinicians	Clinic patients	Clinicians	Patients
Core Purpose	Evaluate results & scheduling	Multi-provider portal	Health tracking & data hub	AI transcription	Visit summaries	AI scribe & note generation	Unified patient record
Data Scope	Epic network only	Connected providers	Device + limited EHRs	Provider EHRs	Local EHR only	Multi-EHR transcription	Mutli-EHR via FHIR APIs
AI features	Basic insights	Limited	Health trends	AI note creation	None	Ambient transcription	Minimal (data sync)
Record Control	Provider	Provider	User (limited)	Provider	Provider	Provider	Patient
Sharing	Within network	Within network	Manual export	EHR only	Staff messaging	With EHR	Controlled patient sharing
Offline Access	Unavailable	Unavailable	Partial	Unavailable	Unavailable	Unavailable	Syncing
Security	HIPAA	HIPAA	Device encryption	HIPAA	HIPAA	HIPAA	HIPAA, FHIR complaint
Strength	Strong hospital ties	Multi-provider use	Fitness + data sync	Reduces clinician burden	Practice messaging	Speeds documentati on	Broad EHR access
Limitation	Epic-only	Network-limited	Shallow clinical scope	Not patient centric	No unification	Provider-only	No AI assistant

The table shows that most existing health platforms focus on either patients or providers but not both, and they often have limited data integration or AI capabilities. MediVault can stand out by offering a unified, AI-powered platform that connects multiple EHRs and gives patient full control of their records.

4. Differentiation

MediVault differentiates itself from existing healthcare record solutions through a unique combination of patient-controlled data aggregation, AI-powered assistance, and segment-specific features that address unmet needs across our target markets. While competitors like MyChart, FollowMyHealth, and Apple Health offer partial solutions, none provide the comprehensive, patient-centric approach that MediVault delivers.

Our platform stands apart through true cross-system unification under complete patient control. Unlike MyChart, which operates only within the Epic ecosystem, or FollowMyHealth, which is limited to its provider network, MediVault aggregates records from multiple EHR systems via FHIR APIs while giving patients full ownership and control of their health data. Apple Health attempts similar aggregation but suffers from shallow clinical scope and requires manual provider connections. OneRecord offers multi-EHR access but lacks AI-powered assistance and still requires users to manually connect to each provider's account. MediVault automates this process and adds intelligent summarization, making it accessible even to non-technical users like our interviewed senior, Eleanor Harris (interview notes in Appendix), who emphasized she does not want "another password nightmare."
The second key differentiator is our AI assistant for real-time clinical documentation. While AI transcription tools like Nuance DAX and Sunoh.ai exist, they serve providers rather than patients. MediVault is the only patient-facing solution that captures doctor visits in real-time through Visit Mode, automatically translating medical jargon into plain language, updates medication lists, flags potential drug interactions, and creates actionable summaries. This directly addresses the core pain point identified of our mental health student interviewees who reported being emotionally drained from repeatedly retelling their medical history to new providers. Our solution positions us uniquely between consumer health apps (which are too shallow) and provider documentation tools (which are not patient-centric), filling a critical gap that generates immediate value for both patients and their doctors.

Our segment-specific features represent our third major differentiator. Through deep customer research involving 30 interviews across patients, caregivers, providers, and insurers, we identified distinct needs that we address through tailored functionality. For mental health patients like Jordan Rivera, we offer granular privacy controls that let users choose exactly what information to share with which providers, plus integrated mood, and sleep tracking alongside therapy notes. For parents like Melissa Chang (interview notes in Appendix), we provide a digital vaccination passport with automated reminders and daycare-ready sharing links that solve the exact problem that caused enrollment delays and near-duplicate vaccinations in our interviews. For seniors and caregivers like Eleanor Harris (interview notes in Appendix), we deliver a large-font interface, offline access capability, and HIPAA-compliant caregiver delegation that allows family members to assist without technical complexity. No existing solution offers this level of segment customization while maintaining a unified platform architecture.

5. Why Now?

The convergence of regulatory mandates, technological maturity, pandemic-driven behavioral shifts, and growing patient empowerment creates an unprecedented window of opportunity for MediVault to succeed where previous attempts at healthcare record unification have failed.

i. Regulatory mandates are forcing interoperability.

The 21st Century Cures Act, implemented in 2020 and fully enforced by 2023[xiii], requires healthcare providers to make patient health information electronically accessible through standardized APIs. The FHIR (Fast Healthcare Interoperability Resources) standard has matured and been widely adopted[xiv], meaning that for the first time in history, technical barriers to cross-system data exchange have been eliminated. Medicare's interoperability mandates (2023+) further accelerate this trend[xv], particularly benefiting our senior segment. Previous attempts at patient record consolidation failed because providers had no obligation to share data, and no technical standards existed. Now, legal, and technical infrastructure supports our business model.

ii. AI and natural language processing have reached clinical viability

Recent breakthroughs in large language models enable accurate real-time transcription and medical terminology translation that were impossible just three years ago[xvi]. Our AI assistant can now reliably convert doctor-patient conversations into structured, actionable summaries without requiring specialized medical vocabulary training. This technological maturity allows us to deliver the Visit Mode feature that differentiates MediVault from static record aggregators. The timing is critical, attempting this even two years ago would have resulted in unacceptably high error rates that would undermine patient trust.

iii. COVID-19 permanently shifted healthcare delivery and patient expectations

The pandemic accelerated telemedicine adoption by 5-7 years, with virtual visits increasing dramatically. More importantly, COVID-19 exposed the dangerous consequences of fragmented records when patients could not access their vaccination history, medication lists, or test results across different providers. According to Fierce Healthcare, 83% of patients were forced to provide duplicate health information at multiple appointments during this period, causing delays in urgent care[xvii]. This widespread frustration has created massive demand for solutions like MediVault that patients previously might have considered unnecessary

iv. Mental health destigmatization drives adoption in our largest segment.

Mental health utilization among young adults has increased dramatically post-pandemic, with 38% of undergraduates experiencing moderate to severe depression symptoms and 61% of college students who need therapy actively seeking it[xviii].Universities are investing heavily in mental health infrastructure, creating institutional partnership opportunities. High prevalence of mental illness among adults aged 18-25 (approximately 31% any mental illness, 2023)[xix] combined with high smartphone ownership (approximately 96% among adults aged 18-25[xx]) creates ideal adoption conditions. Simultaneously, the normalization of digital mental health tools like BetterHelp and Talkspace has overcome previous resistance to managing mental health digitally. Our interviews confirmed that college students are digitally native, privacy-conscious, and frustrated by fragmented records, the perfect early adopter segment.

v. The infant health record crisis reached a tipping point.

Despite decades of digitization efforts, paper vaccination cards remain standard practice in pediatrics, leading to widespread problems that our parent interviews documented. The 2022 infant formula shortage and ongoing daycare enrollment challenges have made parents acutely aware of how fragile paper-based systems are. Boston specifically saw a decline in birth rates but increased focus on preventive pediatric care[xxi], creating demand for better tracking tools. CDC's recommendation of 10+ pediatric visits before age two [xxii]creates multiple pain points that MediVault directly addresses. Parents are ready for a digital solution that eliminates lost cards and missed appointments.

vi. The aging population is creating caregiver crisis conditions.

By 2025, adults aged 65+ represent 17.3% of the U.S. population[xxiii], with Boston's senior population at 84,000 and growing. [xxiv]More critically, 53 million Americans now provide unpaid caregiving[xxv], and 38% of seniors take five or more prescriptions simultaneously[xxvi]. Our interviews revealed that medication errors due to fragmented records are common and dangerous. The combination of growing senior population, increasing polypharmacy complexity, and strained caregiver resources makes MediVault's caregiver access features essential rather than optional. Previous generations managed with paper because they saw fewer specialists; modern geriatric care complexity demands digital coordination.

vii. Smartphone penetration has finally reached critical mass across all segments.

While younger demographics achieved 90%+ smartphone ownership years ago, our target segments have only recently crossed adoption thresholds that make mobile-first solutions viable. Smartphone ownership among seniors reached 61% in 2024, up from under 40% in 2018[xxvii]. Parents under 40 exceed 81% smartphone usage. This infrastructure prerequisite that did not exist five years ago now enables MediVault to reach all three customer segments through a single mobile platform.

viii. Competitive landscape validation proves market readiness.

The entry of tech giants like Apple into health records aggregation (Apple Health launched records feature in 2018) and the recent $50M+ funding rounds for digital health platforms [xxviii] demonstrate investor confidence and market validation. However, these well-funded competitors have focused on breadth rather than depth, creating the opening for MediVault's segment-specific approach. Their partial success has educated the market about the value proposition while their limitations have created frustrated users ready to switch to a more comprehensive solution.

The convergence of these eight factors, regulatory support, technological maturity, pandemic-driven behavior change, mental health normalization, pediatric record crisis, caregiver strain, universal smartphone access, and competitive validation, creates a once-in-a-generation opportunity. Previous attempts at patient record unification faced regulatory barriers, technical limitations, or insufficient user demand. Today, all prerequisites align simultaneously, making this the optimal moment to launch MediVault.

Use Cases

1. Young and Middle-aged Adults Managing Complex Mental-Health Care

Jay, a 29-year-old professional with anxiety, uses MediVault to maintain consistency with his medication and therapy regimen. The application gently prompts him with a brief emotional check-in consisting of two questions every morning. Jay does not need to take notes during treatment because Visit Mode automatically records important conversation points and summarizes his therapist's suggestions, such as breathing techniques, journaling prompts, and goals for the next session. The app is intricately connected to his psychiatrist's prescription updates, which modify reminders and notify him if a pattern of missing doses emerges. Over time, MediVault helps Jay identify triggers and improvements by visualizing his mood and sleep patterns alongside therapy notes. A discreet "Help Now" icon links him to pre-selected emergency contacts and helplines in times of need. MediVault ensures Jay's mental health remains structured, consistent, and secure, helping him stay in control of his progress while safeguarding his privacy and well-being.

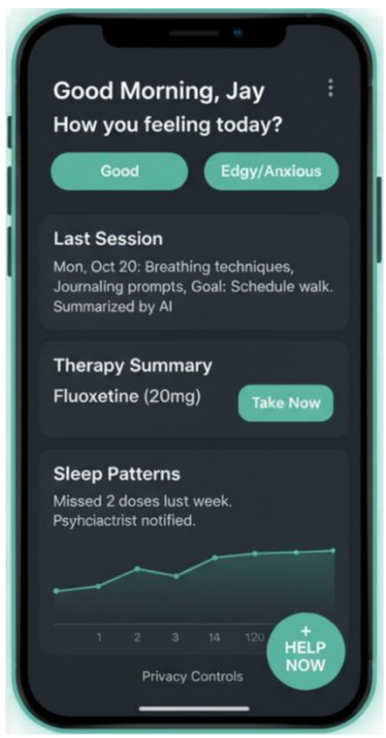

Figure 3 - MediVault app mockup HELP NOW

2. Parents of Infants (0-2 Years)

First-time parents, Rohan and Ananya, must manage their jobs with the regular pediatric visits of their 8-month-old child. When they launch MediVault, the app shows Aanya's vaccination schedule with the next doses indicated. Visit Mode collects the doctor's recommendations for feeding, rashes, and developmental milestones at the clinic. MediVault immediately updates the database with next-due dates as soon as the nurse records Aanya's vaccinations. Once at home, the app automatically schedules pediatric checkups, growth and development phases, and upcoming vaccines. Additionally, it displays Aanya's growth curve in comparison to WHO guidelines, highlighting any anomalies. Ananya shares a read-only link straight from the app when childcare asks for medical information. By combining growth tracking, vaccination tracking, and medical record sharing into a single, easy-to-use platform, MediVault makes parenting easier and gives working parents the confidence and peace of mind they need to manage their child's health with ease.

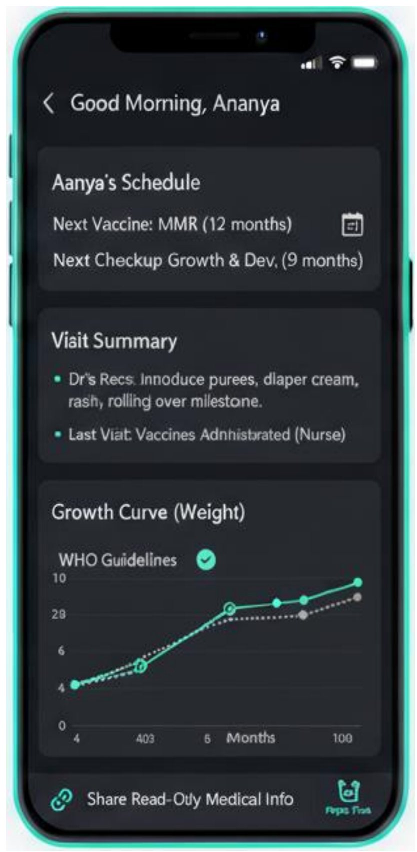

Figure 4 - MediVault app mockup helping young parents to monitor their kid's growth

3. Seniors with Polypharmacy & Caregiver Support

Lalitha, a 72-year-old retiree, chooses to live alone while managing her arthritis and hypertension. MediVault automatically creates a straightforward, large-font summary of her latest vitals, medication regimen, and lab results prior to her regular doctor's appointment. She turns on Visit Mode at the clinic, which listens to the consultation and converts the doctor's advice into straightforward terms. Lalitha is given a detailed rundown of dietary guidelines and dosage adjustments when she gets home. MediVault stores the new prescription in its secure database, automatically updates her medication reminders, and indicates any drug conflicts. She does not require technical assistance to monitor her progress because her lab results are automatically synchronized once she uploads it in MediVault. Through concise summaries, intelligent reminders, and simple record organization, MediVault enables senior users like Lalitha to easily manage their health on their own, fostering self-reliance, safety, and confidence in their everyday care.

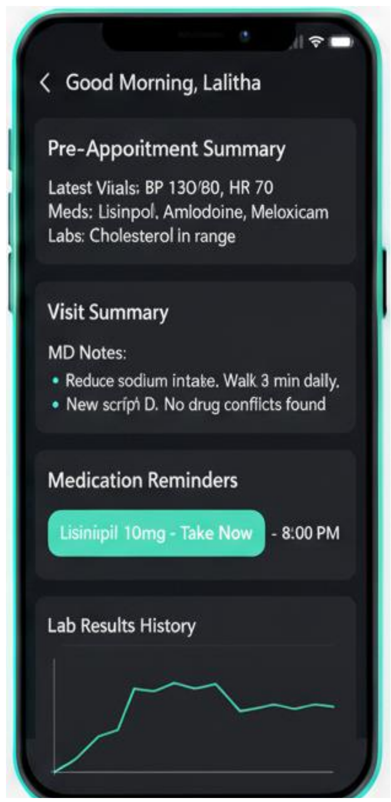

Figure 5 - MediVault app mockup reminding medications intakes and appointments summaries

Market Size

The market sizing analysis for MediVault, a unified personal health-record management and coordination app, was developed based on secondary data and structured assumptions reflecting the three core customer segments:
(A) Young adults managing complex mental-health care[xxix],
(B) Parents of infants (0-2 years)[xxx], and
(C) Seniors with polypharmacy and caregiver support[xxxi].
The analysis estimates the Total Addressable Market (TAM) and Market Potential for an initial launch in the Boston metropolitan area, a representative early-adoption market with a digitally literate healthcare ecosystem[xxxii].

1. Assumptions and Target Population

The total addressable population in Boston was estimated at 500,000 potential users, representing the combined reach across the three core segments:[xxxiii]
This figure was derived by mapping national prevalence data from the CDC**,** NIMH, and U.S. Census Bureau to the Boston metro population distribution.

Segment	**Estimated Population (Boston Metro)**	**Share of Total**
A- Young Adults Managing Mental-Healthcare	200,000	40%[xxxiv]
B – Parents of Infants (0-2 Years)	25,000	5%[xxxv]
C – Seniors with Polypharmacy + Caregiver Support	275,000	55%[xxxvi]
Total Addressable Users	**≈ 500,000**	**100%**

Additional assumptions were made to model engagement and pricing:

- Average subscription price: $8 per month based on comparable digital health apps such as MyChart Plus and CareZone[xxxvii].
- Active usage rate within addressable base: 1 %, 2 %, and 5 % adoption scenarios, consistent with early-stage SaaS benchmarks[xxxviii].
- 12-month revenue cycle per user.
- Gross margin: 60%, aligned with typical digital health SaaS cost structures[xxxix].

2. TAM and Market Potential Calculations

Adoption Scenario	Users	Annual Revenue ($8 × 12 months)	Estimated Profit (60 %)
1 % Adoption	5,000	$480,000 / year	$288,000 / year
2 % Adoption	10,000	$960,000 / year	$576,000 / year
5 % Adoption	25,000	$2.4 million / year	$1.44 million / year

3. Interpretation

At a moderate 2% adoption rate, MediVault could achieve 10,000 paying users, generating $960,000 in Annual Recurring Revenue (ARR) and $576,000 in annual profit.
An aggressive 5% scenario, achievable through provider partnerships and regional healthcare integrations, would yield $2.4 million ARR with $1.44 million profit potential.

4. Conclusion

In conclusion, the market sizing analysis demonstrates that MediVault has a strong, quantifiable opportunity within the combined population of young adults managing mental-health care, parents of infants, and seniors with polypharmacy needs.
The Boston pilot market alone presents a feasible path to $2 million-plus in annual recurring revenue at moderate adoption levels.
These findings underscore the importance of building partnerships with mental-health clinics, pediatric networks, and senior-care programs, which collectively address the most acute coordination of pain points in health-record management.

Caveats / Risks / Key Dependencies

Category	Risk / Failure Mode	Description	Possible Mitigants
1. Privacy & Security	Data breaches or unauthorized access	MediVault will store sensitive medical data. Any breach or mishandling of health information could severely damage user trust and violate HIPAA and other privacy regulations.	• Implement HIPAA and GDPR compliance from inception. • Use end-to-end encryption, multi-factor authentication, and zero-knowledge architecture. • Conduct regular penetration testing and third-party audits.
2. Regulatory Compliance	Complex healthcare data laws (HIPAA, HITECH, GDPR, etc.)	Navigating U.S. and global healthcare data regulations is complex and resource intensive.	• Engage legal counsel specialized in health tech. • Adopt standardized compliance frameworks (e.g., HITRUST, SOC 2). • Start pilot deployment within one U.S. jurisdiction first.
3. Interoperability	Limited access to hospital EHR systems	Many EHRs (Epic, Cerner, etc.) have closed ecosystems, limiting MediVault's ability to pull records automatically.	• Leverage open APIs via the 21st Century Cures Act. • Partner with EHR vendors or health data intermediaries. • Begin with systems offering FHIR-standard APIs.
4. Trust & Adoption	User hesitation due to privacy or complexity concerns	Patients may be reluctant to link all records or trust AI with sensitive health data.	• Offer transparent data ownership (user-controlled storage). • Use plain-language privacy explanations. • Introduce educational content and clinician endorsements.
5. Technology Reliability	AI assistant accuracy and uptime	Errors in transcription, summarization, or data syncing could reduce reliability and trust.	• Continuous model testing and validation. • Offer user verification and edit controls. • Maintain human-in-the-loop support for critical functions.
6. Legal / IP Risks	Potential overlap with existing patented AI documentation tools (e.g., Nuance DAX)	Certain AI audio summarization methods may be covered by existing patents.	• Conduct IP due diligence. • Focus on patient-facing record management features (distinct positioning). • Use open or licensed speech-to-text APIs.
7. Third-Party Dependencies	Reliance on cloud services and API integrations	MediVault depends on external APIs, EHR connections, and secure cloud hosting for functionality. Service outages or policy changes could disrupt access.	• Diversify API and cloud vendors. • Establish data caching and offline fallback. • Use service-level agreements (SLAs) with providers.
8. Financial Sustainability	High development and compliance costs before reaching scale	Achieving HIPAA compliance, API access, and AI reliability may require significant upfront investment.	• Begin with MVP targeting one user segment. • Seek healthcare innovation grants or partnerships. • Introduce tiered pricing to build early revenue.
9. Market Competition	Dominant players could expand features	Established systems could integrate similar functionality before MediVault scales.	• Differentiate through patient ownership and AI summarization. • Focus on underserved cross-provider patients and caregivers. • Build strong UX and advocacy partnerships.

1. Strategic Considerations

The analysis of customer pain points, adoption drivers, and stakeholder requirements reveals primary strategic pillars necessary for MediVault's successful market entry and adoption: Trust & Compliance, Data Capture Strategy, Phased Market Entry, and Sustained Engagement.

i. Trust, Compliance, and Transparency as Core Foundation

MediVault's foundation must be built on visible privacy and security controls, so users feel confident sharing sensitive health data. Features such as audit trails, access logs, and user-controlled sharing turn compliance into a competitive advantage. This strong privacy posture is essential given global and US regulatory requirements and the link between trust and health-data sharing.[xl]

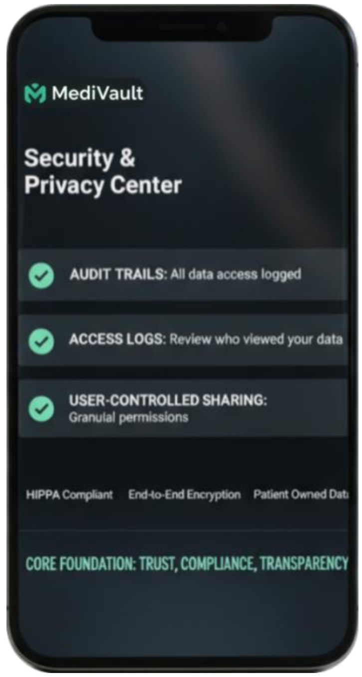

Figure 6 - MediVault app mockup relying on Trust, Compliance and Transparency

ii. AI Visit Mode as a Data Capture Engine

Rather than waiting for full EHR integrations, MediVault's "Visit Mode" should function as a first-mile capture tool that records and structures medical visits, prescriptions, and notes. This lets the product start delivering value early while building toward full interoperability and richer data use.[xli]

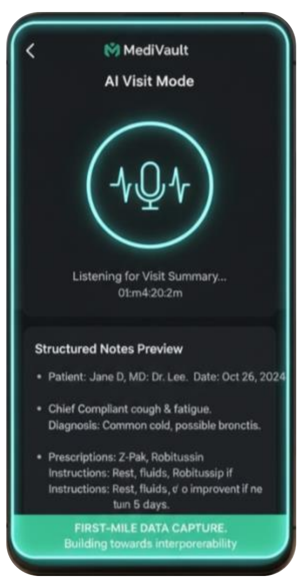

Figure 7 3- *MediVault desktop mockup offering access control*

iii. Unified Timeline and Intelligent Summaries

By aggregating all visits, tests, and prescriptions into a single timeline with AI-generated plain language summaries, MediVault transforms passive storage into a tool that patients and providers actively use. This enhances usability and meaningful engagement.[xlii]

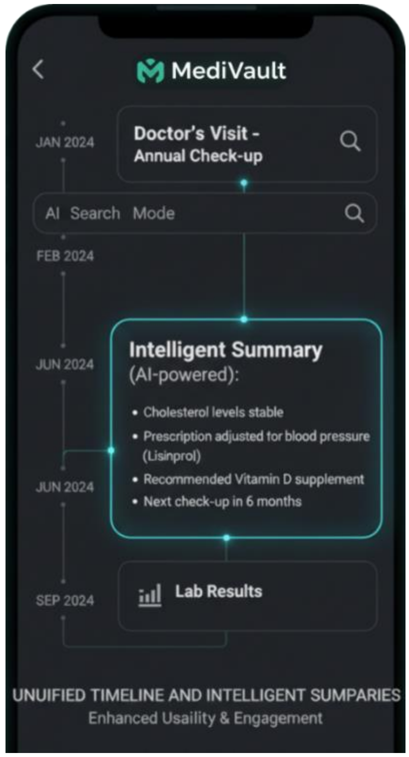

Figure 8 - MediVault app mockup summarizing power

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iv. Adaptive Access Controls for Caregivers and Multi-Stakeholder Use

Different users (seniors, parents, caregivers, clinicians) need different permission levels and access modes. Embedding role-based sharing and read-only links increases value and adoption across segments, enabling meaningful collaboration without sacrificing data control.[xliii]

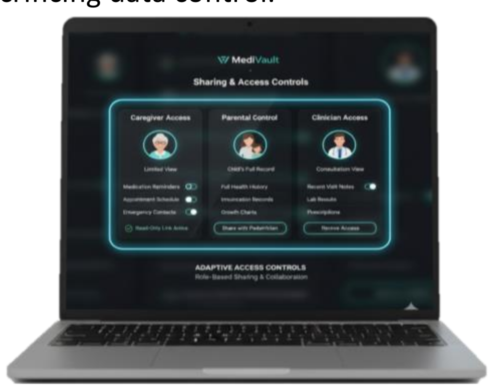

Figure 9 4- MediVault desktop mockup offering access control

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v. Cross-Device Continuity and Offline Reliability

Healthcare data must be accessible anytime, anywhere, anyplace, even when connectivity is poor. MediVault must support offline caching, device sync, and seamless continuity to serve underserved users and build trust in reliability and availability.[xliv]

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vi. Personalization Through Behavioral Insights to Drive Engagement

Beyond storing records, MediVault should use AI to personalize reminders, surface trends (missed doses), and adapt based on user behavior. Beyond storing records, MediVault should use AI to personalize reminders, surface trends (missed doses), and adapt based on user behavior. Engagement is a strategic lever for retention and lifetime value in digital health

Figure 10 - MediVault app mockup

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vii. Phased Market Rollout and Ecosystem Partnerships

Launch initially to digitally savvy young adults to validate core flows and features, then expand to parents and seniors with strategic partnerships (clinics, EHR vendors, insurers). A phased approach improves adoption, builds credibility, and scales sustainability.[xlv]

2. Team Members

i. Founding Team

MediVault was founded by a cross-functional team of graduate innovators from Northeastern University, combining expertise in product management, business strategy, user experience, and healthcare systems. The team's shared mission is to make medical record management intuitive, interoperable, and patient-owned, while building a scalable, AI-assisted platform with minimal technical overhead.

Name	Title / Role	Key Responsibilities
Aditya Girish Anand	Head of Product Strategy	Defines product vision and roadmap, leads MVP feature prioritization, and ensures alignment with regulatory and clinical requirements.
Dorian Chirackal	Technical Strategy Lead	Oversees technology stack selection, interoperability framework (FHIR/HL7) integration, and AI-driven backend automation.
Karishma Bahl	UX Design Lead	Designs user interfaces for patients and caregivers, focusing on accessibility, simplicity, and HIPAA-compliant design.
Matthieu Delatour	Business Operations Lead	Drives go-to-market strategy, partnership development, financial modeling, and investor relations.
Sangeetha Phaneesh	Research & Validation Lead	Conducts stakeholder and user research (patients, clinics, insurers) and synthesizes adoption insights to inform product direction.

ii. Lean Expansion Plan (Post-MVP Growth)

The founding team intends to operate under a "lean + AI-augmented" model, minimizing engineering and operations costs while maintaining enterprise-grade output. The goal is to rapidly validate market demand, achieve interoperability milestones, and prepare for pilot deployment with healthcare partners.

Proposed Role	Strategic Purpose	AI/Automation Support
AI-Augmented Full Stack Engineer (Contract)	Build the MVP using low-code tools and AI-assisted code generation to reduce traditional development costs.	GitHub Copilot, OpenAI API, Bubble/FlutterFlow
Compliance & Security Advisor (Part-Time)	Guide HIPAA, GDPR, and interoperability compliance, ensuring data security and audit readiness.	AI-driven policy and risk scan tools
Growth & Marketing Strategist	Launch automated digital campaigns and manage early-stage growth channels with AI analytics.	HubSpot AI, Jasper, Notion AI
Clinical Partnership Manager	Establish collaborations with clinics and insurers for pilot testing and data onboarding.	CRM automation and AI-enabled outreach
AI Systems Specialist (Fractional)	Oversee integration of generative AI features for summarization, patient support, and record insights.	LLM-based workflow automations

iii. Team Readiness Summary

The MediVault founding team brings a complementary mix of strategic, technical, and user-centered expertise across healthcare and digital product management. With a shared commitment to leveraging AI to reduce costs and accelerate development, the team is well-positioned to execute a focused, capital-efficient roadmap from MVP to pilot deployment. Their balance of research rigor, operational discipline, and market awareness provides a strong foundation for scalable growth and investor confidence.

3. Go/No Go Recommendation

Recommendation: **GO** (Proceed to MVP Development and Launch)

User surveys and industry studies confirm a significant unmet need for a secure, unified, patient-controlled health record solution. The market need for MediVault is validated by the current landscape:

1. **High Pain Point:** The problem of fragmented health data (validated by studies showing up to 83% of patients repeatedly providing duplicate information) is acute and directly impacts quality, safety, and efficiency of care.
2. **Strong Differentiator:** MediVault stands apart from traditional hospital portals with its patient-owned data model and AI Assistant that automatically records and summarizes medical visits. This transforms it from a passive aggregator into an intelligent, action-driven platform for managing personal health data.
3. **Favorable Adoption Drivers:** High smartphone ownership, growing trust in digital health tools, and increasing demand for transparency make the market highly receptive to MediVault's launch.

Conditions for **GO** (Phase 1: MVP Development - Segment A: Young Adults Managing Complex Mental Health Care)

1. **Security Certification:** Establish and fund a clear path to HIPAA compliance and third-party audit readiness before development begins
2. **Feature Discipline:** Limit the MVP to three validated core functions: Unified Health Timeline, AI Visit Mode (recording and summarization), and Prescription Reconciliation, while deferring non-essential features to later releases.
3. **User Experience Focus:** Conduct UX testing specifically with Segment A (Young Adults Managing Complex Mental Health Care**)** to ensure AI capture, intuitive navigation, and trust-centered data presentation that reduces administrative burden.

Meeting these conditions will ensure MediVault launches as a compliant, user-validated, and scalable digital health platform ready for pilot deployment.

Appendix: Notes from Interviews

1. Buyers

Q1: Tell me about how you or your caregiver currently keep track of your medical records, prescriptions, and appointments. What works well and what's frustrating?

Q2: Can you describe a time when missing or incomplete information caused confusion, delays, or mistakes in your care? How did it make you feel?

Q3: Have you ever lost trust in a digital health system? What specifically happened, and what would make you feel confident using one?

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i. Segment A: Young Adults Managing Complex Mental-Health Care

Interview #1

Age 25, Jack Logan, Security Manager at University of Philadelphia

Q1: Tell me about how you currently keep track of your therapy notes, medications, and appointments. What

works well and what is frustrating?
A1: “I keep therapy notes scattered across my phone's notes app and in various emails from my counselor. It works in the moment because I can jot things down quickly, but trying to find specific information later is inconvenient. Everything is disorganized across different apps and folders, so I waste time searching when I need to reference something. I wish there was one central place where everything was organized and easy to access.”
Q2: Can you describe a time when missing or incomplete information caused by confusion, delays, or mistakes in your care? How did it make you feel?
A2: “When I switched to a new counselor, they had no access to my previous therapy notes or history. I had to spend the entire first session retelling everything from scratch—my background, symptoms, what we'd been working on. It felt like such a waste of time and money, and honestly it was emotionally exhausting to rehash everything again. I was frustrated that there wasn't a better way to transfer that information.”
Q3: Have you ever lost trust in a digital health system? What specifically happened, and what would make you feel confident using one?
A3: “I'd trust a digital system if I could control exactly who sees what information—like granting and revoking access to specific providers. Privacy is key for me, especially with mental health records. I need to know that my information isn't visible to anyone unless I explicitly approve it. Without that level of control, I wouldn't feel comfortable putting my therapy notes in any digital system.”

Interview #2

Age 21, Emily Chen at Northeastern University
Q1: Tell me about how you currently keep track of your therapy notes, medications, and appointments. What works well and what is frustrating?
A1: “I mostly rely on my memory to keep track of therapy appointments and what medications I'm taking. It's simple because there's nothing to maintain, but honestly, I sometimes forget important details about medications or things my therapist told me. There's no backup if I can't remember something which has caused problems. I know it's not a reliable system, but I haven't found anything better that I trust.”
Q2: Can you describe a time when missing or incomplete information caused by confusion, delays, or mistakes in your care? How did it make you feel?
A2: “I once forgot to mention a prescription I'd taken in the past, and my doctor almost prescribed the same medication again. When I realized mid-conversation, it was alarming because I could have ended up with duplicate prescriptions or wasted money. I felt irresponsible for not keeping better records but also frustrated that there's no easy way to track this. It made me realize how risky my memory-based system really is.”
Q3: Have you ever lost trust in a digital health system? What specifically happened, and what would make you feel confident using one?
A3: “If the app is officially verified or endorsed by my school's health center or my healthcare provider, I'd definitely consider using it. Otherwise, I just don't trust random health apps with my sensitive mental health information. I need some kind of institutional backing or certification to feel confident. Without that credibility, I'd worry about data breaches or my information being misused.”

Interview #3

Age 23, Rahul Mehta at Boston University, Law School
Q1: Tell me about how you currently keep track of your therapy notes, medications, and appointments. What works well and what is frustrating?
A1: “I use my campus counseling portal to track appointments and some basic notes from sessions there. It's

convenient for on-campus care, but it doesn't include anything from my outside therapy sessions or psychiatrist visits. Having an incomplete picture of my mental health care makes it frustrating and not very useful. I end up having to remember or track the outside appointments separately anyway."

Q2: Can you describe a time when missing or incomplete information caused by confusion, delays, or mistakes in your care? How did it make you feel?

A2: "I've had to explain my entire depression history and treatment background repeatedly to different providers because nothing transfers between systems. Each time I see someone new, it's like starting from zero again. It's emotionally exhausting to keep reliving and explaining the same difficult experiences, and it makes me feel like I'm not making progress. I just wish the information could follow me instead of having to retell my story repeatedly."

Q3: Have you ever lost trust in a digital health system? What specifically happened, and what would make you feel confident using one?

A3: "I want to see clear, straightforward privacy policies that explain exactly how my data is used and protected. If the language is vague or filled with legal loopholes, I absolutely won't use it. As a law student, I know how important fine print is, and I need transparency about data sharing and storage. Clear policies would give me confidence that my rights are being respected."

Interview #4

Age 20, Sophia Martinez at University of Michigan, MS of Mechanical Engineering

Q1: Tell me about how you currently keep track of your therapy notes, medications, and appointments. What works well and what is frustrating?

A1: "I write therapy notes and medication information on paper in a notebook because it feels private and secure. The problem is I often lose the notebook or leave it at home when I need it at appointments. It's completely unreliable when I'm moving between my dorm, classes, and the health center. I end up rewriting things or trying to remember what I wrote down, which defeats the purpose."

Q2: Can you describe a time when missing or incomplete information caused confusion, delays, or mistakes in your care? How did it make you feel?

A2: "My doctor asked for records from previous appointments and old prescription information, and I couldn't find them anywhere. I'd either lost the papers or left them in my dorm room across campus. It was really embarrassing to show up unprepared, and I felt stressed because the appointment couldn't be as productive without that information. I looked disorganized even though I'd been trying to keep track of things."

Q3: Have you ever lost trust in a digital health system? What specifically happened, and what would make you feel confident using one?

A3: "I'd trust a digital system if I could permanently delete my own records myself anytime I want. I need to know that if I decide I don't want something stored anymore, I can remove it immediately without having to contact support or wait. That level of control over my own data would make me feel much more comfortable. Without deletion rights, I'd worry about things staying in the system forever."

Interview #5

Age 24, Daniel Carter at Stanford University, PhD

Q1: Tell me about how you currently keep track of your therapy notes, medications, and appointments. What works well and what is frustrating?

A1: "I track all my medications, dosages, and schedule changes in a detailed spreadsheet on my laptop. It's thorough and well-organized when I'm at home, but it's not convenient to carry everywhere I go. I can't always access it on my phone, and I can't bring my laptop to every appointment. The lack of portability means the information isn't available when I need it most."

Q2: Can you describe a time when missing or incomplete information caused confusion, delays, or mistakes in your care? How did it make you feel?

A2: "I missed an important dosage update my psychiatrist had made because I didn't have my spreadsheet with me at a follow-up appointment with my primary care doctor. They were working with outdated information, which could have caused problems with medication interactions. It was stressful to realize the gap in communication, and I felt frustrated with myself for not having a better system. I knew my spreadsheet method had limitations, but this really highlighted them."

Q3: Have you ever lost trust in a digital health system? What specifically happened, and what would make you feel confident using one?

A3: ". If the system uses strong encryption and requires password protection, maybe even biometric authentication. I'd trust it with my health information. I need to know the data is secure both in transit and at rest, not just sitting on a server somewhere. Technical security measures are more important to me than vague promises about privacy. If I can verify the encryption standards, I'd feel confident using it."

Interview #6

Age 25, Aisha Khan at University of Texas, Austin, Data Analytics Professional

Q1: Tell me about how you currently keep track of your therapy notes, medications, and appointments. What works well and what is frustrating?

A1: "I depend almost entirely on my memory to track therapy appointments, medications, and what we discuss in sessions. It's simple because there's no system to maintain, but it's not reliable, especially during stressful times when my memory isn't great. I've forgotten appointment times and medication details more than once. I know I need a better system, but I haven't found one I'm comfortable with yet."

Q2: Can you describe a time when missing or incomplete information caused confusion, delays, or mistakes in your care? How did it make you feel?

A2: "I forgot to mention a potential medication interaction to my doctor because I couldn't remember all the supplements I was taking. It ended up causing some uncomfortable side effects that could have been avoided. I was really frustrated with myself for not keeping better records, and it made me feel irresponsible about my own health care. The experience made me realize how dangerous relying on memory can be."

Q3: Have you ever lost trust in a digital health system? What specifically happened, and what would make you feel confident using one?

A3: ". I don't trust big tech companies with my personal health data after all the privacy scandals I've read about. But if a healthcare provider I already seek my university health center or my therapist, officially endorsed and used the app, I might be willing to try it. I need institutional trust and accountability to feel safe. Without a trusted organization backing it, I'd assume my data would be sold or misused."

Interview #7

Age 25, Michael O'Neill at New York University Medical School

Q1: Tell me about how you currently keep track of your therapy notes, medications, and appointments. What works well and what is frustrating?

A1: "I have therapy notes in one app, medication lists in another, and appointment reminders in my phone's calendar. It's fragmented across multiple places, which makes it hard to find specific information quickly when

I need it. Each app has its own interface and search function, so I often can't remember where I saved something. The lack of integration means I'm constantly switching between apps and still missing things."
Q2: Can you describe a time when missing or incomplete information caused confusion, delays, or mistakes in your care? How did it make you feel?
A2: "I once had to repeat the same blood test because I couldn't find my previous results when the doctor asked for them. I knew I'd had the test done recently, but the results were buried in an email or app somewhere I couldn't locate. It was annoying to have to redo it, wasteful of time and money, and frustrating to know it was completely preventable. I felt like my disorganized system was literally costing me."
Q3: Have you ever lost trust in a digital health system? What specifically happened, and what would make you feel confident using one?
A3: "Trust comes from complete transparency about data handling. I want to know exactly where my data is stored, who has access, and how it's being used. Vague privacy policies or hidden data-sharing agreements are dealbreakers for me. If a system can show me a clear audit trail and explicit consent for every use of my data, I'd trust it. Without that transparency, I'd assume the worst about how my information is being handled."

Interview #8

Age 25, Sarah Patel at Greenstone Capital, MBA in Finance
Q1: Tell me about how you currently keep track of your therapy notes, medications, and appointments. What works well and what is frustrating?
A1: "Everything related to my health care is on paper, prescription lists, therapy notes, appointment summaries, all of it. My husband and I try to bring the relevant papers to appointments, but we sometimes forget them at home or in the car. It's bulky to carry around, and we don't always know which papers we'll need ahead of time. The paper system feels secure but is impractical for our busy schedules."
Q2: Can you describe a time when missing or incomplete information caused confusion, delays, or mistakes in your care? How did it make you feel?
A2: "The doctor couldn't confirm which vaccinations had already been administered because we forgot to bring our records to the appointment. They almost gave me a vaccine I'd already received, which would have been unnecessary and potentially harmful. It was stressful to realize we'd almost had a medical error because of missing paperwork. I felt foolish for not being more organized with something so important.
Q3: Have you ever lost trust in a digital health system? What specifically happened, and what would make you feel confident using one?
A3: "I'd trust a digital system if it comes directly from a reputable hospital, medical association, or pediatric organization that I already know and trust. It would also need to have secure access controls so I can manage exactly who can view the information. Institutional credibility combined with strong security features would give me confidence. Without both of those elements, trustworthy source, and proven security. I wouldn't risk putting my family's health data online."

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ii. Segment B: Parents of Infants (0-2 Years)

Interview #9

Age 40, James Howard at AeroMotion Systems, MS in Mechanical Engineering, father of a 6-year-old
Q1: Tell me about how you or your caregiver currently keeps track of your child's medical records,

vaccinations, and appointments. What works well and what is frustrating?
A1: "We track vaccines in a calendar at home, just writing down dates when shots are due. It's simple and doesn't require any apps or technology, which I appreciate. But honestly, it's easy to miss updates when life gets busy, especially if the calendar isn't in a visible spot. I wish there was a way to get automatic reminders without having to manage it all manually."
Q2: Can you describe a time when missing or incomplete information caused confusion, delays, or mistakes in your child's care? How did it make you feel?
A2: "Our pediatrician asked for records from when my son was younger that we didn't have with us. We had to reschedule the appointment and dig through old paperwork at home. It caused delays in getting him the care he needed and created unnecessary stress for our whole family. I felt frustrated that something so important could slip through the cracks so easily"
Q3: Have you ever lost trust in a digital health system? What specifically happened, and what would make you feel confident using one?
A3: "I'd only trust a digital system if parents have full control over who can access the data and how it's shared. I need to be able to approve every request and revoke access instantly if needed. Without that level of control, I wouldn't feel comfortable putting my child's sensitive health information online. Transparency about data usage would also be essential for me to trust any platform."

Interview #10

Age 32, Laura Gomez at PeopleFirst Solutions, MA in Human Resources Management, mother of a 2-year-old
Q1: Tell me about how you or your caregiver currently keeps track of your child's medical records, vaccinations, and appointments. What works well and what is frustrating?
A1: "I keep everything in a big three-ring binder, vaccination cards, appointment summaries, growth charts, everything. It feels secure having physical copies, but the binder has gotten so thick and disorganized over time. When I need to find something specific, like a particular test result, I must flip through dozens of pages. It's messy and not efficient when I'm in a hurry at the doctor's office."
Q2: Can you describe a time when missing or incomplete information caused confusion, delays, or mistakes in your child's care? How did it make you feel?
A2: "We had a referral to a specialist, and they needed our daughter's complete vaccination history and some lab results. I couldn't find all the paperwork in my binder, so we showed up incompletely. The appointment ended up being partly wasted because they couldn't move forward without the full records, and I felt so anxious and unprepared as a parent."
Q3: Have you ever lost trust in a digital health system? What specifically happened, and what would make you feel confident using one?
A3: "If the data is properly encrypted and I have a guarantee that it won't be sold to third parties or advertisers, I'd be okay using a digital system. I've heard too many stories about health data being misused for marketing purposes. I need clear, legally binding privacy policies that are enforced. If those protections are in place and verified, I'd consider switching from my binder."

Interview #11

Age 25, Kevin Wu at CloudForge Technologies, BS in Computer Science, father of a 3-year-old
Q1: Tell me about how you or your caregiver currently keeps track of your child's medical records, vaccinations, and appointments. What works well and what is frustrating?
A1: "We rely mainly on our pediatric clinic's patient portal to access records and schedule appointments. It's

convenient when it works, but the information there is often incomplete or takes days to update after visits. Sometimes vaccination dates don't match what's on our paper card, which creates confusion about what's been administered. The inconsistency makes me question whether I can really depend on it."

Q2: Can you describe a time when missing or incomplete information caused confusion, delays, or mistakes in your child's care? How did it make you feel?

A2: "The portal failed to send us a reminder about an upcoming vaccination, and we completely missed the scheduled appointment. We only realized it when we got a follow-up call weeks later asking why we hadn't come in. It was stressful because we pride ourselves on staying on top of our son's health, and this made us feel like we'd dropped the ball as parents."

Q3: Have you ever lost trust in a digital health system? What specifically happened, and what would make you feel confident using one?

A3: "Trust for me depends on the system having legitimate government or medical institution backing; not just some random app created by a startup. I'd want to see partnerships with recognized healthcare organizations or regulatory approval. Without that kind of credibility and accountability, I wouldn't risk putting my child's health information into the system, no matter how convenient it claims to be."

Interview #12

Age 38, Priya Nair at Sunrise Family Clinic, MSN in Nursing Practice, mother of a 4-year-old

Q1: Tell me about how you or your caregiver currently keeps track of your child's medical records, vaccinations, and appointments. What works well and what is frustrating?

A1: "I use WhatsApp to share pictures of vaccine cards and appointment notes with my spouse so we're both informed. It's convenient because we both have the app and can access photos anytime. But over time, finding specific information means scrolling through hundreds of chat messages, which is completely disorganized. There's no structure or way to categorize things, so it becomes chaotic quickly."

Q2: Can you describe a time when missing or incomplete information caused confusion, delays, or mistakes in your child's care? How did it make you feel?

A2: "One time, I sent the daycare a photo of my daughter's vaccination card, but it was too blurry for them to read properly. They rejected it and said they needed a clear, legible copy before she could attend that day. I had to leave work, drive home to get the original card, scan it properly, and send it again—all while dealing with frustrated daycare staff and a crying child."

Q3: Have you ever lost trust in a digital health system? What specifically happened, and what would make you feel confident using one?

A3: "I'd trust an app if it maintained detailed access logs showing exactly who viewed my child's data and when they accessed it. I'd also need the ability to control permissions granularly deciding who sees what information and for how long. As a nurse, I understand how important audit trails are for accountability in healthcare. Those features would give me the confidence that nothing is happening with the data without my knowledge."

Interview #13

Age 31, Anna Lee at BrightCare Clinic, Nurse Practitioner, mother of a 2-year-old

Q1: Tell me about how you or your caregiver currently keeps track of your child's medical records, vaccinations, and appointments. What works well and what is frustrating?

A1: "We use the standard hospital vaccination card that they gave us at birth, and we keep it in a folder at home with other important documents. It works fine most of the time, and I like having a physical record. But

the card is small and easy to misplace, especially when we're traveling or moving things around the house. Sometimes we need it urgently and can't find it, which creates last-minute panic."
Q2: Can you describe a time when missing or incomplete information caused confusion, delays, or mistakes in your child's care? How did it make you feel?
A2: "We misplaced our daughter's vaccination card right before daycare enrollment was supposed to start. The daycare had a strict policy requiring proof of vaccinations before the first day, and they wouldn't make any exceptions. We spent hours frantically searching the house and calling the clinic for a replacement, causing absolute panic. The whole situation could have derailed our childcare plans and work schedules."
Q3: Have you ever lost trust in a digital health system? What specifically happened, and what would make you feel confident using one?
A3: ". I'd need strong assurance that my child's health data isn't shared with anyone (schools, insurance companies, researchers) without my explicit, informed consent. Every single request should require my approval first. I've seen too many cases where data gets used for purposes parents never intended or agreed to. Clear opt-in policies and the ability to revoke consent at any time would be non-negotiable for me."

Interview #14

Age 34, David Park at BrightWave Media, BA in Marketing, father of a 3-year-old
Q1: Tell me about how you or your caregiver currently keeps track of your child's medical records, vaccinations, and appointments. What works well and what is frustrating?
A1: "I keep notes in a baby tracking app about feeding and sleep schedules, but medical information stays on paper cards from the doctor's office. It feels fragmented because I must check two completely different places depending on what information I need. The app doesn't integrate with any medical systems, so I'm manually entering appointment dates and vaccine names. It's redundant work and leaves room for errors when I'm tired or distracted"
Q2: Can you describe a time when missing or incomplete information caused confusion, delays, or mistakes in your child's care? How did it make you feel?
A2: "I forgot to bring my son's growth chart to a routine pediatric visit because it was in a different folder at home. The doctor needed to compare current measurements to previous ones, so they had to redo several measurements they'd already done at the last visit. It felt like such a waste of time and made me look disorganized and unprepared as a parent, which was embarrassing and stressful."
Q3: Have you ever lost trust in a digital health system? What specifically happened, and what would make you feel confident using one?
A3: "I'd trust a digital system if it had two-factor authentication for every login and crystal-clear access control settings. I want to see exactly who has permission to view my son's information and be able to change those permissions easily. Coming from a marketing background, I'm very aware of how data can be exploited, so I need proof that security is taken seriously with multiple layers of protection."

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iii. Segment C: Seniors with Polypharmacy & Caregiver Support

Interview #15

Age 68, Margaret Collins, Retired Teacher, BA in Education
Q1: Tell me about how you or your caregiver currently keep track of your medical records, prescriptions, and appointments. What works well and what is frustrating?
A1: "I mostly keep track of my prescriptions and doctor notes in a small notebook and store pill bottles in a cabinet organized by day. I've tried digital options, but they always feel too complicated to keep up with. My

system is old-fashioned, but at least I can see everything in one place; though it's easy to forget something when switching doctors."

Q2: Can you describe a time when missing or incomplete information caused confusion, delays, or mistakes in your care? How did it make you feel?

A2: "Once, a new doctor almost prescribed me a medication that conflicted with something I was already taking because I forgot to bring my list. It made me realize how fragile my system is; one missing note could impact my health. That incident still makes me nervous before every visit."

Q3: Have you ever lost trust in a digital health system? What specifically happened, and what would make you feel confident using one?

A3: "I get scared hearing about data breaches in the news, so I never felt comfortable trusting apps with my medical history. If I ever do use one, I'd want to see clear proof of strong security and no ads or third-party data use."

Interview #16

Age 72, George Thompson, Retired Engineer, MS in Civil Engineering

Q1: Tell me about how you or your caregiver currently keep track of your medical records, prescriptions, and appointments. What works well and what is frustrating?

A1: "My wife and I keep paper folders for each specialist we see. Each visit gets its own section; lab reports, scans, prescriptions, everything. It works fine until a new doctor need something from another clinic, and then we're left scrambling to find or fax it."

Q2: Can you describe a time when missing or incomplete information caused confusion, delays, or mistakes in your care? How did it make you feel?

A2: "Once, the hospital lost track of my blood test results, and I had to redo the entire thing. It wasn't just frustrating that my treatment delayed and cost extra money. That was the moment I wished all my records were in one place."

Q3: Have you ever lost trust in a digital health system? What specifically happened, and what would make you feel confident using one?

A3: "I trust digital systems only if my family can access and manage them for me. I've seen too many confusing portals and forgotten passwords. A system that lets a trusted caregiver help would make a big difference."

Interview #17

Age 70, Helen Brooks, Retired Nurse, BSN in Nursing

Q1: Tell me about how you or your caregiver currently keep track of your medical records, prescriptions, and appointments. What works well and what is frustrating?

A1: "I keep a mix of paper notes and occasional emails to my doctors. Since I worked in healthcare, I try to stay organized, but managing multiple specialists gets messy. Each provider seems to use a different system, so I end up being the messenger."

Q2: Can you describe a time when missing or incomplete information caused confusion, delays, or mistakes in your care? How did it make you feel?

A2: "I once forgot to mention a drug allergy during a consultation, and it led to a reaction that could've been avoided. That experience made me realize how important complete, accessible records are, especially as we age and see more doctors."

Q3: Have you ever lost trust in a digital health system? What specifically happened, and what would make you feel confident using one?

A3: "I'd trust a digital platform only if it's very secure, something with PIN protection, no ads, and preferably tied to a known hospital system. Anything commercial feels risky."

Interview #18

Age 45, Anita Desai, Caregiver, daughter of 78-year-old patient Meera Desai

Q1: Tell me about how you or your caregiver currently keep track of your medical records, prescriptions, and appointments. What works well and what is frustrating?

A1: "I manage all my mom's medical records in a big binder, separated by doctor and date. It takes time, but at least I know where everything is. I've tried using hospital portals, but most don't talk to each other, so I still must print and carry everything around."

Q2: Can you describe a time when missing or incomplete information caused confusion, delays, or mistakes in your care? How did it make you feel?

A2: "Once, two specialists prescribed clashing medications because one didn't have access to the other's notes. It was scary and stressful, especially since my mom's condition is fragile. That experience showed me how disconnected healthcare really is."

Q3: Have you ever lost trust in a digital health system? What specifically happened, and what would make you feel confident using one?

A3: "Trust for me means having control. I'd only use a digital tool if I could decide exactly who has access to what, and revoke it anytime. Most apps don't make that easy or transparent."

Interview #19

Age 74, Robert Miller, Caregiver for spouse Linda Miller, Age 71

Q1: Tell me about how you or your caregiver currently keep track of your medical records, prescriptions, and appointments. What works well and what is frustrating?

A1: "We keep everything in a file cabinet with folders for every doctor and prescription. It's a lot of paper, but I don't trust uploading sensitive medical data online. Still, it's stressful to manage, especially when we have emergency appointments."

Q2: Can you describe a time when missing or incomplete information caused confusion, delays, or mistakes in your care? How did it make you feel?

A2: "During one emergency, I couldn't find Linda's latest test results, and the ER team had to redo them. It caused unnecessary panic and delay in her care. That's when I realized how fragile a paper-only system really is."

Q3: Have you ever lost trust in a digital health system? What specifically happened, and what would make you feel confident using one?

A3: "If an app worked offline and was recommended by our doctor's office, I'd consider it. I just don't want something that requires constant internet or sends data somewhere I can't see."

Interview #20

Age 66, Evelyn Parker, Retired Librarian, MA in Library Science

Q1: Tell me about how you or your caregiver currently keep track of your medical records, prescriptions, and appointments. What works well and what is frustrating?

A1: "I document everything in a small notebook I bring to each visit. It's my version of a health diary, notes about medications, symptoms, and next appointments. It's reliable until I forget the notebook at home, which happens more than I'd like."

Q2: Can you describe a time when missing or incomplete information caused confusion, delays, or mistakes in your care? How did it make you feel?
A2: "Once, I forgot the notebook before a cardiology checkup, and the doctor couldn't verify my dosage history. It caused confusion and made me feel embarrassed. That day I wished there was a safe, digital version I could always access."
Q3: Have you ever lost trust in a digital health system? What specifically happened, and what would make you feel confident using one?
A3: "I'd trust a system only if it was created and endorsed by my hospital network. I've seen too many third-party apps come and go, and I wouldn't risk losing my data or privacy."

2. Sellers

Q1: How do you currently manage patient records and share information with external systems or other providers? What are the main challenges you face in interoperability?

Q2: What factors influence your decision to adopt a new patient-facing digital health tool or record consolidation platform? What concerns or barriers prevent adoption?

Q3: What requirements or safeguards would make you confident that a digital system meets privacy, security, and regulatory standards? How do you evaluate trust in such systems?

i. Clinics / Hospitals

Interview #1

Age: 52, Dr. Lisa Carter, Chief Medical Officer, Boston General Hospital
Q1: How do you currently manage patient records and share information with external systems or other providers? What are the main challenges you face in interoperability?
A1: "We mostly rely on our Epic EHR system for managing patient records within our hospital network. When we need to share records with smaller clinics or providers using different EHR systems, the process becomes incredibly tedious and often requires manual faxing or sending files through secure email. The lack of standardized interfaces means we spend significant administrative time on what should be automated transfers. It's frustrating because interoperability was supposed to be solved by now, but in practice it remains a major bottleneck."
Q2: What factors influence your decision to adopt a new patient-facing digital health tool or record consolidation platform? What concerns or barriers prevent adoption?
A2: "Security and compliance are absolutely critical; we cannot compromise on those fronts given the regulatory environment. We also need comprehensive staff training programs and a clear return on investment before we'll commit resources to new technology. Tools that add to our workload or duplicate existing workflows rarely get adopted, no matter how innovative they claim to be. Our staff is already stretched thin, so anything new must genuinely streamline processes rather than create additional administrative burden."
Q3: What requirements or safeguards would make you confident that a digital system meets privacy, security, and regulatory standards? How do you evaluate trust in such systems?
A3: "HIPAA compliance is non-negotiable—that's the baseline requirement for any system we consider. We also require independent third-party security audits, end-to-end encryption for data in transit and rest, and detailed audit trails that log every single access to patient records. Additionally, we strongly prefer tools that

come recommended by reputable healthcare IT organizations or industry consortiums we trust. Vendor reputation and their record with other major healthcare institutions heavily influence our evaluation process."

Interview #2

Age: 48, Mark Thompson, Director of IT, Cambridge Health Clinic
<u>Q1</u>: How do you currently manage patient records and share information with external systems or other providers? What are the main challenges you face in interoperability?
<u>A1</u>: "We use an EHR system that allows limited API connections, which helps in some cases but not comprehensively. Sharing records across different health systems often fails completely due to mismatched data formats and incompatible standards. Even when the technical connection exists, the data doesn't always translate correctly between systems. We end up having to manually reformat or re-enter information, which defeats the purpose of having integrated systems in the first place."
<u>Q2</u>: What factors influence your decision to adopt a new patient-facing digital health tool or record consolidation platform? What concerns or barriers prevent adoption?
<u>A2</u>: "The number one requirement is that it needs to integrate seamlessly with our existing EHR without disrupting current workflows. If a new tool creates duplicate data entry or requires staff to maintain information in two places, we simply won't use it. Our clinicians are already overwhelmed with EHR documentation requirements, so adding another layer would be counterproductive. The integration must be bidirectional and automatic, not something that requires manual synchronization."
<u>Q3</u>: What requirements or safeguards would make you confident that a digital system meets privacy, security, and regulatory standards? How do you evaluate trust in such systems?
<u>A3</u>: "Encrypted connections using current best practices are essential, we need to see TLS 1.3 or better for all data transmission. Role-based access control is crucial so that staff only see information relevant to their role, and we need the ability to revoke access instantly when someone leaves or changes positions. We also specifically check for SOC 2 Type II and HITRUST certifications, as those demonstrate a vendor has undergone rigorous independent audits. Without these certifications, a vendor won't even make it to the consideration phase."

Interview #3

Age: 55, Dr. Amanda Lee, Chief of Internal Medicine, North Shore Hospital
<u>Q1</u>: How do you currently manage patient records and share information with external systems or other providers? What are the main challenges you face in interoperability?
<u>A1</u>: "We primarily use Cerner as our EHR system across the hospital. Exchanging records with other hospitals, even those also using Cerner, is surprisingly slow and error-prone due to configuration differences. Sometimes results arrive as non-structured PDFs, which means the data isn't actionable within our EHR and staff must manually extract information. It's particularly problematic for lab results and imaging reports that should integrate directly into our clinical decision support tools but instead arrive as static documents."
<u>Q2</u>: What factors influence your decision to adopt a new patient-facing digital health tool or record consolidation platform? What concerns or barriers prevent adoption?
<u>A2</u>: "Integration with existing clinical workflows is absolutely key, if it doesn't fit naturally into how our physicians and nurses already work, it won't be adopted. Clinical validation of the data is also crucial because we worry significantly about liability if information in the system is inaccurate or incomplete. If a provider makes a clinical decision based on faulty data from a third-party platform, the liability questions become very complex. We need assurances about data quality and clear protocols for handling discrepancies or errors."

Q3: What requirements or safeguards would make you confident that a digital system meets privacy, security, and regulatory standards? How do you evaluate trust in such systems?
A3: "We require recognized compliance certifications like HIPAA, HITECH, and HITRUST to even begin the conversation. Secure cloud hosting with clear data residency policies—we need to know exactly where our data is stored geographically—is essential. Regular penetration testing by qualified third parties gives us confidence in the security posture. Honestly, recommendations from peer institutions matter tremendously; if similar-sized hospitals we respect are successfully using a platform, that carries more weight than any vendor pitch."

Interview #4

Age: 47, Sarah Jenkins, Director of Patient Services, Cambridge Family Clinic
Q1: How do you currently manage patient records and share information with external systems or other providers? What are the main challenges you face in interoperability?
A1: "We use an EHR portal that allows our patients to access some of their records and schedule appointments. But when patients switch to different providers or see specialists outside our network, their records aren't automatically shared—we still rely on faxes or emailing PDFs back and forth. It's antiquated and time-consuming, and patients often get frustrated waiting for records to be transferred. The lack of interoperability means patients sometimes receive duplicate tests or have treatment delays while we wait for information."
Q2: What factors influence your decision to adopt a new patient-facing digital health tool or record consolidation platform? What concerns or barriers prevent adoption?
A2: "Ease of use is crucial for both our staff and our patients—if either group finds it confusing or cumbersome, adoption will fail. If a new platform requires significant extra administrative work from our front desk or medical assistants, it won't be adopted regardless of its other benefits. Our team is already managing heavy workloads, and we can't add responsibilities without taking something else away. The system needs to actively reduce administrative burden, not just promise to do so in the future."
Q3: What requirements or safeguards would make you confident that a digital system meets privacy, security, and regulatory standards? How do you evaluate trust in such systems?
A3: "We need comprehensive HIPAA-compliance documentation that we can review and audit ourselves, not just a vendor's assurance. Detailed audit logs that show who accessed what information and when are essential for both security and addressing patient concerns. Clear data ownership rules must be established patients need to understand they own their data and can control access. We also require the technical ability to revoke system access immediately if there's a security concern or contract termination."

Interview #5

Age: 50, Dr. Kevin O'Connor, Pediatrician, Harborview Children's Clinic
Q1: How do you currently manage patient records and share information with external systems or other providers? What are the main challenges you face in interoperability?
A1: "We rely heavily on paper vaccination cards that we hand out to parents, supplemented by records in our EHR system. Digital sharing with parents is quite limited—we can give them portal access, but many don't use it consistently. We often receive incomplete immunization records when children transfer from other practices or states. The fragmentation means we sometimes can't verify what vaccines a child has received, which creates both medical and liability concerns for our practice."
Q2: What factors influence your decision to adopt a new patient-facing digital health tool or record consolidation platform? What concerns or barriers prevent adoption?

A2: “Any new tool must demonstrably improve patient care without increasing our clinical or administrative workload—that's the bottom line. Parental engagement and empowerment are important goals, but they absolutely cannot come at the expense of security or patient privacy. We need parents to have access to their children's records, but with appropriate safeguards and audit trails. If a platform helps parents stay organized without creating extra work for our staff, we'd be very interested.”
Q3: What requirements or safeguards would make you confident that a digital system meets privacy, security, and regulatory standards? How do you evaluate trust in such systems?
A3: “Two-factor authentication for all user access is a must—passwords alone aren't sufficient protection anymore. Comprehensive audit trails that log every access, modification, and sharing event give us both security and accountability. Secure APIs with proper authentication and authorization mechanisms are essential if the system connects to our EHR. Vendor reputation is critical; we need to see a proven record with other pediatric practices and no history of data breaches or compliance violations.”

Interview #6

Age: 53, Michael Brown, IT Manager, Greenfield Medical Center
Q1: How do you currently manage patient records and share information with external systems or other providers? What are the main challenges you face in interoperability?
A1: “We manage everything through Epic EHR, which works well within our organization and with other large Epic users. However, sharing records with smaller independent clinics that use different systems is often manual and incredibly time-consuming. We end up printing to PDF, faxing, or using secure email—methods that feel decades old. The irony is that we've invested millions in our EHR, but interoperability still requires staff to manually manage record transfers like it's 1995.”
Q2: What factors influence your decision to adopt a new patient-facing digital health tool or record consolidation platform? What concerns or barriers prevent adoption?
A2: “Cost is always a factor—we need to justify any new technology investment to our board with clear benefits. Integration complexity is a major barrier; if implementation will take months and require extensive customization, that's a deterrent. Staff training costs and time are often underestimated but represent real expenses. Anything that duplicates existing workflows or requires maintaining parallel systems is an absolute non-starter; our staff won't tolerate redundant data entry under any circumstances.”
Q3: What requirements or safeguards would make you confident that a digital system meets privacy, security, and regulatory standards? How do you evaluate trust in such systems?
A3: “Encrypted connections using industry-standard protocols for all data transmission are mandatory. We need regular compliance reports and documentation that we can review and present to our board and auditors. Vendor support history matters—we want to see evidence of responsive customer service and a record of addressing security issues promptly. Peer recommendations from other healthcare IT managers are very influential; we're part of several regional health IT consortiums where we share vendor experiences and would heavily weight those testimonials.”

Interview #7

Age: 46, Dr. Priya Singh, Chief Medical Officer, Boston Children’s Outpatient Clinic
Q1: How do you currently manage patient records and share information with external systems or other providers? What are the main challenges you face in interoperability?
A1: “We use multiple different EHR systems across our departments, which creates internal interoperability challenges even within our own clinic. Lab results often arrive late or to the wrong inbox, delaying clinical

decisions. Parents frequently bring partial or photocopied information from other clinics that it’s difficult to verify or integrate into our system. The fragmentation means we can't always get a complete picture of a child's health history, which impacts the quality of care we can provide.”
Q2: What factors influence your decision to adopt a new patient-facing digital health tool or record consolidation platform? What concerns or barriers prevent adoption?
A2: “If a platform genuinely streamlines communication between providers and reduces medical errors, we are interested. However, liability concerns loom large—if we make clinical decisions based on incorrect or incomplete data from an external platform, who bears the responsibility? We need clear contractual protections and data quality guarantees. The system must have robust error-checking and validation to ensure information accuracy, or we can't rely on it for clinical decision-making.”
Q3: What requirements or safeguards would make you confident that a digital system meets privacy, security, and regulatory standards? How do you evaluate trust in such systems?
A3: “Regular independent security audits, give us confidence that security measures are being maintained over time. Full HIPAA compliance with documented policies and procedures is obviously required. The ability to restrict access by role is crucial; our receptionists shouldn't have the same access level as our physicians. We also need granular permission controls so we can customize access based on specific job functions and clinical responsibilities.”

Interview #8

Age: 49, Dr. Robert Lewis, Primary Care Physician, Riverside Clinic
Q1: How do you currently manage patient records and share information with external systems or other providers? What are the main challenges you face in interoperability?
A1: “Our EHR has very limited interoperability capabilities, which is frustrating given how expensive these systems are. We routinely receive incomplete patient histories from outside providers, leaving gaps in our understanding of a patient's medical background. This forces us to either make clinical decisions with partial information or delay treatment while we track down missing records. The inefficiency impacts both patient satisfaction and clinical outcomes, yet there's no easy solution given the fragmented healthcare IT landscape.”
Q2: What factors influence your decision to adopt a new patient-facing digital health tool or record consolidation platform? What concerns or barriers prevent adoption?
A2: “Time savings for both staff and patients is the key value proposition we look for in any new tool. If staff members must spend more time learning, maintaining, or working around a new system, it's not worth it regardless of its theoretical benefits. Anything requiring extra manual data entry will almost certainly fail—our providers are already drowning in EHR documentation requirements. We need automation and actual workflow improvement, not just digitization of existing paper processes.”
Q3: What requirements or safeguards would make you confident that a digital system meets privacy, security, and regulatory standards? How do you evaluate trust in such systems?
A3: “Independent third-party security audits provide objective validation that we can trust. Strong encryption for data both in transit and at rest is non-negotiable in today's threat environment. Vendor transparency about their security practices, incident response procedures, and any past breaches is essential; we need to know we're working with honest partners. A vendor who tries to hide or minimize security issues loses our trust immediately.”

Interview #9

Age: 51, Sarah Mitchell, Director of Nursing, Bayview Medical Center
Q1: How do you currently manage patient records and share information with external systems or other

providers? What are the main challenges you face in interoperability?
A1: “We mostly use our EHR supplemented by scanned documents for anything that arrives on paper. Patients often bring outdated records from other providers that may not reflect their current health status. We must verify everything which takes nursing time that could be spent on direct patient care. The mix of digital and paper creates inefficiencies and increases the risk of missing important information buried in scanned documents that aren't searchable."
Q2: What factors influence your decision to adopt a new patient-facing digital health tool or record consolidation platform? What concerns or barriers prevent adoption?
A2: “. Any new tool must integrate smoothly with our current EHR system, we can't manage multiple disconnected platforms. Staff acceptance is crucial; if the nurses and medical assistants don't embrace it, adoption will fail no matter what leadership wants. Most importantly, it cannot slow down patient care or add steps to our clinical workflows. We operate on tight schedules, and any system that adds even 30 seconds per patient would be unacceptable when multiplied across hundreds of daily encounters.”
Q3: What requirements or safeguards would make you confident that a digital system meets privacy, security, and regulatory standards? How do you evaluate trust in such systems?
A3: “HIPAA-compliant cloud hosting with clear data protection standards is mandatory, we need to know our data is being handled properly. Role-based access control ensures that each staff member only sees information necessary for their job function. Comprehensive audit trails that track every access and modification give us both security monitoring and the ability to investigate any concerns. These aren't nice-to-have features; they're absolute requirements for any system handling patient data.”

Interview #10

Age: 54, Dr. Helen Roberts, Director of Health IT, Metro Health Clinic
Q1: How do you currently manage patient records and share information with external systems or other providers? What are the main challenges you face in interoperability?
A1: “We operate multiple EHR systems across different departments, which creates internal data silos. Data sharing between systems is extremely limited and often requires us to fall back on faxing or creating PDFs. The lack of integration means important clinical information doesn't flow seamlessly between providers within our own organization, let alone with external partners. It's embarrassing to admit that our sophisticated digital systems sometimes perform worse than paper records did in terms of information accessibility.”
Q2: What factors influence your decision to adopt a new patient-facing digital health tool or record consolidation platform? What concerns or barriers prevent adoption?
A2: “Staff workload is always our first consideration—will this make their jobs easier or harder? Patient usability matters because low adoption rates will undermine any system's value. We need clear, quantifiable ROI projections, not vague promises of future benefits. Anything that adds complexity to our already complicated IT environment is a non-starter; we're looking to simplify, not add another layer. Our staff is experiencing burnout, so new tools must provide obvious, immediate value or they'll be resisted.”
Q3: What requirements or safeguards would make you confident that a digital system meets privacy, security, and regulatory standards? How do you evaluate trust in such systems?
A3: “Encrypted data storage using current cryptographic standards is foundational. Recognized compliance certifications like SOC 2, HITRUST, and HIPAA provide third-party validation of security practices. Strong vendor reputation backed by verifiable references from other healthcare organizations is crucial—we want to talk to current customers, not just read marketing materials. A vendor's financial stability also matters; we can't risk implementing a system from a company that might not be around in three years.”

ii. Insurers

Interview #11

Age: 45, Robert Jenkins, VP of Product, United Health Insurance

Q1: How do you currently manage claims and medical records for your insured members? What are the main challenges?

A1: "We receive data from hundreds of providers across the country in various formats—some send HL7 feeds, others send PDFs, some still fax information. The lack of standardized records leads directly to claim delays because our systems can't automatically process non-standard formats. Staff must manually review and enter data, which introduces errors and slows everything down. It's a massive operational inefficiency that costs us millions annually in processing overhead and delayed payments."

Q2: What factors influence your decision to adopt a new digital patient-facing platform? What barriers prevent adoption?

A2: "A new platform must demonstrably reduce claim processing errors, which directly impact both costs and member satisfaction. Full HIPAA compliance is obviously non-negotiable—any breach would be catastrophic for us. The platform needs to integrate seamlessly with our existing claims workflows and adjudication systems. We can't have claims processors working in multiple systems or manually transferring data; everything must flow automatically through our established processes."

Q3: What requirements or safeguards would make you confident in a patient-facing platform?

A3: "End-to-end encryption for all data transmission and storage protects both our members and our business. Secure multi-factor authentication prevents unauthorized access from compromised credentials. Detailed audit trails for every record access allow us to detect anomalies and investigate potential security incidents. These technical controls must be combined with strong vendor governance and regular security assessments to maintain trust over time."

Interview #12

Age: 50, Laura Peterson, Director of Innovation, BlueCross BlueShield

Q1: How do you currently manage claims and medical records for your insured members? What are the main challenges?

A1: "We receive PDFs or various EHR exports from providers, but the standardization is poor across the industry. Different providers structure information differently, use different coding systems, and include varying levels of detail. This leads to delays processing while staff manually review submissions and often results in denied claims when documentation is insufficient. The lack of structured, standardized data is one of the biggest operational challenges we face in claims processing."

Q2: What factors influence your decision to adopt a new digital patient-facing platform? What barriers prevent adoption?

A2: "Interoperability with provider EHR systems is critical—we need data to flow automatically rather than through manual processes. Patient adoption rate matters because a platform is only valuable if people use it consistently. Security is non-negotiable; any platform handling PHI must meet the highest security standards. We also need to see evidence that the platform improves outcomes—faster claim processing, fewer errors, better member satisfaction—before committing to adoption."

Q3: What requirements or safeguards would make you confident in a patient-facing platform?

A3: "HIPAA-compliant hosting infrastructure with appropriate business associate agreements is the foundation. Strong encryption protocols protect data from unauthorized access or breaches. Clearly defined data ownership rules must establish that members own their health data and we're merely custodians. We also need contractual provisions allowing us to audit the vendor's security practices and requiring prompt notification of any security incidents."

Interview #13

Age: 48, Michael Davis, Claims Manager, Cigna
Q1: How do you currently manage claims and medical records for your insured members? What are the main challenges?
A1: “We work with multiple provider networks nationwide, and each has different documentation standards and submission processes. Incomplete documentation is a persistent problem that delays claim approvals and frustrates both providers and members. We often have to send requests for additional information, which extends the cycle time and increases administrative costs. The fragmentation across provider systems makes it nearly impossible to achieve the processing efficiency we're striving for.”
Q2: What factors influence your decision to adopt a new digital patient-facing platform? What barriers prevent adoption?
A2: “The platform must streamline claims processing, not create more manual work for our already busy claims teams. Data accuracy is essential—if we're making payment decisions based on incorrect information, it creates financial and legal liability. The system needs to validate data quality and flag inconsistencies automatically. If it just digitizes existing problems without solving them, there's no value proposition for us.”
Q3: What requirements or safeguards would make you confident in a patient-facing platform?
A3: “End-to-end encryption using industry-standard algorithms protects sensitive health and financial information. Comprehensive audit logs allow us to track who accessed what information and when, which is crucial for compliance and investigation. Clear vendor compliance documentation, including SOC 2 reports, penetration testing results, and security certifications—gives us confidence in their security posture. We need vendors who take security as seriously as we do.”

Interview #14

Age: 52, Jennifer White, Director of Digital Health, Anthem
Q1: How do you currently manage claims and medical records for your insured members? What are the main challenges?
A1: “Data arrives from providers in various formats—HL7, CDA, PDF, even faxed documents—which creates significant challenges. This leads to manual reconciliation work where staff must normalize data into our systems. The inconsistency increases processing time, introduces errors, and makes it difficult to analyze trends across our member population. We've invested heavily in automation, but it can only work effectively with standardized, structured data inputs.”
Q2: What factors influence your decision to adopt a new digital patient-facing platform? What barriers prevent adoption?
A2: “The platform must integrate bidirectionally with both our claims system and provider EHR systems—data needs to flow both ways automatically. It must be secure and reliable with minimal downtime; claim processing can't stop because a vendor's system is down. We need guaranteed uptime SLAs and robust disaster recovery capabilities. The system also needs to handle our scale; we process millions of claims monthly, so performance under load is critical.”
Q3: What requirements or safeguards would make you confident in a patient-facing platform?
A3: “Full HIPAA compliance with documented policies, procedures, and staff training is mandatory. Role-based access control ensures that only authorized personnel can access sensitive information based on job function. Strong encryption for data at rest and in transit protects against breaches. We also require regular security assessments and the right to audit vendor practices to ensure ongoing compliance with our security standards.”

Interview #15

Age: 47, David Allen, Health IT Director, Humana
Q1: How do you currently manage claims and medical records for your insured members? What are the main challenges?
A1: “We interface with multiple provider systems, and data quality varies significantly—sometimes it's complete and structured, other times it's missing critical information or poorly formatted. These inconsistencies force manual review and follow-up, delaying claim processing and payment. Members get frustrated when their claims are pending for weeks while we chase down missing information. The lack of data standardization across the healthcare industry is a fundamental problem that impacts every insurer.”
Q2: What factors influence your decision to adopt a new digital patient-facing platform? What barriers prevent adoption?
A2: “A new platform must reduce our administrative burden and error rates—those are directly measurable ROI metrics we care about. It must maintain strict patient privacy protections; any privacy breach would be devastating to our reputation and result in regulatory penalties. The system needs to improve efficiency without requiring massive workflow changes that would necessitate retraining thousands of employees. Gradual, seamless integration is far preferable to disruptive wholesale replacement.”
Q3: What requirements or safeguards would make you confident in a patient-facing platform?
A3: “Data encryption using current best practices protects information from unauthorized access or interception. Detailed audit trails allow us to monitor system usage and investigate any anomalies or potential security incidents. Compliance with both CMS regulations and HIPAA requirements is mandatory—we're heavily regulated and need vendors who understand that environment. Certification by recognized healthcare IT security frameworks like HITRUST provides independent validation.”

Interview #16

Age: 46, Sarah Collins, VP Operations, UnitedHealth Group
Q1: How do you currently manage claims and medical records for your insured members? What are the main challenges?
A1: “Non-standard EHR exports are a constant challenge—each vendor formats data differently, making automated processing difficult. This leads to delays in approvals and payment processing while staff manually extract and enter information. The inefficiency impacts provider satisfaction and cash flow, which damages our relationships with network providers. We need industry-wide data standards, but until that happens, we're stuck with expensive manual workarounds.”
Q2: What factors influence your decision to adopt a new digital patient-facing platform? What barriers prevent adoption?
A2: “Ease of use for both patients and providers is crucial—if either group finds it difficult, adoption will suffer and the platform won't deliver value. The system must improve claims efficiency by reducing processing time and error rates, which directly impact our operational costs. We need measurable improvements, not just theoretical benefits. The platform also needs to scale to handle millions of members and transactions without performance degradation.”
Q3: What requirements or safeguards would make you confident in a patient-facing platform?
A3: “HIPAA compliance is the baseline—we won't even consider non-compliant solutions. Strong encryption protects member data from breaches and unauthorized access. Clear role-based access controls ensure that users only see information necessary for their job function, following least-privilege principles. We also need

the ability to quickly revoke access when employees leave or change roles to prevent unauthorized data exposure."

Interview #17

Age: 49, Thomas Bennett, Director, Digital Health, Aetna

Q1: How do you currently manage claims and medical records for your insured members? What are the main challenges?
A1: "Data from multiple providers is inconsistent in format, completeness, and timing, which causes significant operational challenges. This forces manual follow-ups with providers to obtain missing information or clarify ambiguous documentation. The back-and-forth delays claim resolution and frustrate everyone involved providers want to be paid promptly, members want claims processed quickly, and our staff wants to work efficiently. The lack of standardization across the industry creates friction at every step."
Q2: What factors influence your decision to adopt a new digital patient-facing platform? What barriers prevent adoption?
A2: "The platform must improve claims accuracy—reducing both false positives and false negatives in our adjudication processes. It needs to integrate seamlessly with provider systems so data flows automatically without manual intervention. Maintaining patient privacy is paramount; we're custodians of extremely sensitive information and take that responsibility seriously. Any platform we adopt must have robust privacy controls and transparent data handling practices that members can trust."
Q3: What requirements or safeguards would make you confident in a patient-facing platform?
A3: "Comprehensive audit trails that log every access, modification, and sharing event provide accountability and security monitoring. Strong encryption using modern algorithms protects data throughout its lifecycle. Recognized compliance certifications like SOC 2 Type II and HITRUST demonstrate that independent auditors have verified security controls. We also look for vendor transparency about their security practices, incident response procedures, and any past security events."

Interview #18

Age: 51, Rachel Kim, Senior Manager, Health IT, Blue Shield

Q1: How do you currently manage claims and medical records for your insured members? What are the main challenges?
A1: "Incomplete patient records from providers are a persistent problem that delays claim processing and increases denial rates. We often receive partial documentation that doesn't support the services billed, requiring us to request additional information. This extends the claim lifecycle, delays provider payment, and creates dissatisfaction all around. Better data quality and completeness from the start would dramatically improve our processing efficiency and accuracy.""
Q2: What factors influence your decision to adopt a new digital patient-facing platform? What barriers prevent adoption?
A2: "Integration with existing systems is critical—we can't have isolated platforms that don't communicate with our core infrastructure. Security must be enterprise-grade with multiple layers of protection; we're a major target for cyber-attacks. Patient adoption and engagement are important because the platform only delivers value if people use it. We need to see proof of concept and pilot results before committing to full-scale deployment across our member population."
Q3: What requirements or safeguards would make you confident in a patient-facing platform?
A3: "Full HIPAA compliance with appropriate business associate agreements and regular compliance audits is mandatory. Strong encryption for data transmission and storage protects against breaches and unauthorized

access. Auditable access logs allow us to track who viewed what information and investigate any suspicious activity. We also require vendors to maintain cyber insurance and have incident response plans we can review."

Interview #19

Age: 50, James Reynolds, Director of Claims, Humana
Q1: How do you currently manage claims and medical records for your insured members? What are the main challenges?
A1: "We receive records in multiple formats from thousands of providers—structured data, PDFs, scanned documents, faxes—which creates a processing nightmare. Manual reconciliation is required to normalize everything into our systems, which is time-consuming, expensive, and error-prone. The lack of standardization means we can't fully automate claim adjudication despite significant technology investments. Staff spends far too much time on data entry rather than exception handling and complex claim review."
Q2: What factors influence your decision to adopt a new digital patient-facing platform? What barriers prevent adoption?
A2: "The platform must reduce errors in claim processing, which impact both costs and member satisfaction. It needs to streamline workflows so claims move through adjudication faster without sacrificing accuracy. Maintaining regulatory compliance is non-negotiable—we operate in a heavily regulated environment with severe penalties for violations. Any new system must fit within our existing compliance framework and potentially strengthen it, not create new compliance risks."
Q3: What requirements or safeguards would make you confident in a patient-facing platform?
A3: "Strong encryption for all data—both at rest and in transit—protects against breaches and unauthorized access. Detailed access logs allow us to monitor usage patterns and detect anomalies that might indicate security issues. Full HIPAA compliance with documented policies and regular audits ensures we meet regulatory requirements. We also assess vendor financial stability and business continuity plans to ensure they'll be reliable partners long-term."

Interview #20

Age: 48, Patricia Evans, VP Digital Transformation, Anthem
Q1: How do you currently manage claims and medical records for your insured members? What are the main challenges?
A1: "Records arrive from multiple providers in different formats—some electronic, some still on paper or fax. Data inconsistencies create processing delays while staff manually review and corrects information. Fields might be labeled differently, dates formed inconsistently, or critical information missing entirely. These problems compound across millions of claims annually, creating significant operational inefficiency and cost. Standardization across the industry would solve many of these challenges but remains elusive."
Q2: What factors influence your decision to adopt a new digital patient-facing platform? What barriers prevent adoption?
A2: "Integration with our existing claims and member management systems is essential—we can't operate disconnected platforms. Ease of use matters for both members and our staff; complicated interfaces lead to errors and low adoption. Patient engagement is important because health outcomes improve when people are actively involved in their care. We need platforms that genuinely add value, not just digitize existing problems or create new administrative headaches."
Q3: What requirements or safeguards would make you confident in a patient-facing platform?
A3: "HIPAA compliance is the foundation—any vendor must have robust compliance programs and regular

audits. Strong encryption using industry-standard protocols protects sensitive health and financial information. Comprehensive audit trails provide visibility into who accessed what data and when, supporting both security and compliance. Vendor reputation and references from other major insurers carry significant weight; we want to work with proven, trustworthy partners who understand the healthcare industry's unique requirements.”

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